30 June ESAT predecessor + IAL P2 recall pack

30 June ESAT predecessor + IAL P2 — Reader-safe recall pack

Source and status boundary

Sole content authority: the final revision guide PDF with SHA-256 `d1c96c58d9e672a52cac4c81cfc82188e36833f99081ec98da6ddb7f3fa9515f`. This pack is a fresh, local downstream derivative. It does not reuse legacy cards, raw visual-model material, personal data, or external/private payloads.

How to use the labels. "Completed live" describes work finished during the recorded tutorial. "Completed with prompting" means it was finished with substantial help. "Official-reference only" is correct study material, but was not completed live. Do not turn an official-reference answer into a claim about what happened in the lesson.

The four protected boundaries: NSAA Q19 target 1/45; NSAA Q20 bearing 326°; P2 Q3; and P2 Q8(ii) beyond n=2k+1 are **official-reference only / not completed live**. P2 Q3 was blank because there was no time. Q8(ii) stopped after the clean first line. The NSAA questions are official predecessor questions used for ESAT preparation; their specimen catalogue provenance is mirror-sourced, as declared in the guide.

1. Jargon glossary

Term	Student-friendly meaning	Exact meaning / symbols	What it is NOT	Status-safe lesson connection
Independent events	If one event does not change the probability of the other, an "and" event is found by multiplying.	$P(A \cap B) = P(A)P(B)$ when A, B are independent.	Not an instruction to add probabilities; addition is for appropriate "or" events.	Q19: both dice showing a specified face is an AND event. The target answer is official-reference only.
Probability mass	A probability model must account for every possible result exactly once.	For the biased die, if outcomes 1–5 each have probability p, then $P(6) = 1 - 5p$.	Not a licence to label p as both a single-outcome probability and a total probability.	Q19 tutor point: label the five equal outcomes consistently.
Bearing	A direction measured from north, turning clockwise, and written with three digits where needed.	Bearings are modulo 360°; e.g. $-34^\circ \equiv 326^\circ$.	Not an angle measured from the page's vertical line unless that line is north.	Q20: the final bearing is official-reference only.

Term	Student-friendly meaning	Exact meaning / symbols	What it is NOT	Status-safe lesson connection
Exterior angle	The turn made while walking around a polygon.	A regular n-gon has exterior angle $360^\circ/n$; a pentagon has 72°.	Not the pentagon's interior angle, which is 108°.	Q20 teaching point; use turns to connect headings.
Arithmetic sequence/series	A sequence changes by the same amount each time; a series adds its terms.	$u_n = a + (n-1)d$ and $S_n = \frac{n}{2}[2a + (n-1)d]$.	Not a geometric sequence, whose ratio rather than difference is constant.	Q1 was tutor-confirmed correct; no narrated method was retaught.
Binomial coefficient	The coefficient that counts how many ways a chosen power can arise in an expansion.	In $(a+b)^n$, term $r+1$ is $\binom{n}{r}a^{n-r}b^r$.	Not permission to change the exponent while copying the expression.	Q2: retain exponent 8 before expanding.
Stationary point	A point where the gradient is zero; it still needs classification.	Solve $A'(x) = 0$, then use $A''(x)$ to distinguish a minimum or maximum.	Not automatically a minimum merely because $A' = 0$.	Q3 is entirely official-reference study; no live working exists.
Geometric series	Each term is multiplied by a fixed ratio.	$S_n = a(1 - r^n)/(1 - r)$ and $S_\infty = a/(1 - r)$ for $\lvert r\rvert < 1$.	Not a series where a listed total may be silently treated as a single term.	Q4: 70.2 is S_3, not ar^2; completed with prompting.
Factor Theorem	A linear factor tells you a value of x that makes the polynomial zero.	$(x - c)$ factor of $f(x) \iff f(c) = 0$; $(3x - 4)$ gives $f(4/3) = 0$.	Not the same as a remainder condition.	Q5: "factor" triggers $f(4/3) = 0$.
Remainder Theorem	Dividing by $(x - c)$ leaves remainder $f(c)$.	Remainder on division by $(x - 2)$ is $f(2)$.	Not "the remainder equals 8" before solving; here the stated remainder is parameter b.	Q5 boundary: use $f(2) = b$, then derive $b = 8$.
Perpendicular bisector	The line through a segment's midpoint at right angles to it; every point on it is equally distant from the endpoints.	For non-vertical lines, perpendicular gradients satisfy $m_1m_2 = -1$.	Not a line through an endpoint, and not a line using the midpoint's x-coordinate as its y-coordinate.	Q6: through $M = (5, 4)$ means $y - 4 = -3(x - 5)$.
Natural domain of logarithms	Every log input must be positive before an algebraic candidate is accepted.	For Q7(ii), $5 - a > 0$ and $a + 25 > 0$, hence $-25 < a < 5$.	Not a non-strict endpoint condition, and not a crossed-out candidate with no reason.	Q7: reject 20 because it violates positivity; accept -5.
Counter-example / parity proof	One valid counter-example disproves a universal claim; a parity proof represents every odd integer as $2k + 1$.	For Q8(ii), $n = 2k + 1$, $k \in \mathbb{Z}$, and the target contains n^3.	Not a decimal-length test for irrationality, and not a square expansion for a cube target.	Q8(i) completed with prompting. Q8(ii) is official-reference only after its first line.

2. Formula and case recall table

Quantity / case	Formula or action	Symbols / conditions	When to use	Trap and status-safe check
Q19 biased die	From $\frac{1}{6}q = \frac{1}{18}$, get $q = \frac{1}{3}$; then $P(1_{\text{biased}}) = \frac{1-q}{5} = \frac{2}{15}$. Target: $\frac{1}{6} \cdot \frac{2}{15} = \frac{1}{45}$.	$q = P(6_{\text{biased}})$; total 12 requires both sixes.	A fair and a biased die are independent.	Multiply for AND. Final target $1/45$ was not completed live: official-reference only.
Q20 pentagon heading	Exterior turn $= 360^\circ/5 = 72^\circ$; first leg is two turns before third: $110^\circ - 2(72^\circ) = -34^\circ \equiv 326^\circ$.	Clockwise turns; bearing measured clockwise from north.	A regular-polygon route gives one heading and asks for another.	108° is only the interior angle. 326° is official-reference only.
Q1 arithmetic series	$n = (254-2)/3 + 1 = 85$; $S = \frac{85}{2}(2+254) = 10\,880$.	$a = 2$, $d = 3$, last term 254.	Constant-difference sequence with a known last term.	Use as answer check only: the lesson did not narrate a method.
Q2 first four binomial terms	$(2-5x)^8 = 256 - 5120x + 44800x^2 - 224000x^3 + \cdots$; for 2.05^8, set $2 - 5x = 2.05$, so $x = -0.01$.	Apply $\binom{8}{r}2^{8-r}(-5x)^r$.	A positive-integer binomial expansion and a representation choice.	Preserve exponent 8; “state” asks only for x, not the numerical approximation.
Q3 open cuboid	$3x^2y = 120 \Rightarrow y = 40/x^2$; $A = 3x^2 + 320/x$; $A' = 6x - 320/x^2 = 0$; $x \approx 3.76$; $A'' = 6 + 640/x^3 > 0$.	Positive x; $P = 3$, $Q = 320$.	Optimise an area after eliminating a constrained dimension.	Entire solution is official-reference only: question was blank/no time live.
Q4 geometric series	Put $a/(1-r) = 75$ into $a(1-r^3)/(1-r) = 70.2$: $75(1-r^3) = 70.2$, $r = 2/5$, $a = 45$.	$\lvert r\rvert < 1$ for S_∞.	Both a finite sum and sum to infinity are supplied.	$70.2 = S_3$, not ar^2; completed with prompting.
Q5 factor / remainder / factorise	$f(4/3) = 0 \Rightarrow 16a + 9b = 56$; $f(2) = b \Rightarrow a = -1$; then $b = 8$; $f(x) = (3x-4)(x+2)(x-1)$.	$f(x) = 3x^3 + ax^2 - 10x + b$.	Linear factor and stated remainder conditions.	The given remainder is b, not 8. Use brackets in long-division subtraction.
Q6 bisector and circle	$M = (5,4)$; $m_{AB} = 1/3$, so $m_l = -3$; $y - 4 = -3(x-5)$ gives $y = -3x + 19$; $k = -2$; $r^2 = (11-7)^2 + (6+2)^2 = 80$.	$A = (-1,2)$, $B = (11,6)$, $P = (7,k)$.	A chord’s perpendicular bisector locates the centre.	Correct point-slope term is $y - 4$, not $y - 5$; radius is from P to B.

Quantity / case	Formula or action	Symbols / conditions	When to use	Trap and status-safe check
Q7 trapezium and logs	$\frac{1}{2}\log[4\cdot 9\cdot(5\cdot 6\cdot 7\cdot 8)^2] = \log_{10} 10080$; $(5-a)^2 = 5(a+25) \Rightarrow (a-20)(a+5)=0$.	Strip width 1; log inputs positive.	Convert a weighted sum of logs and solve a log equation.	The $\frac{1}{2}$ becomes a square root. Domain $-25 < a < 5$: reject 20, accept -5.
Q8 counter-example / proof	(i) $\sqrt{2}(3\sqrt{2}) = 6$. (ii) $(2k+1)^3 + 3(2k+1) + 2 = 4(2k^3+3k^2+3k+1)+2 \equiv 2 \pmod 4$.	$k \in \mathbb{Z}$; every odd $n = 2k+1$.	Disprove a universal irrational-product claim; prove parity for all odd integers.	Cube, not square. Q8(ii) beyond $n = 2k+1$ is official-reference only.

3. Formula-booklet boundary

Supplied on the Pearson IAL P2 formula page: arithmetic-series term and sum; geometric-series term, finite sum and sum to infinity; change of base; the binomial theorem for positive integer index; and the trapezium rule.

Learn / derive: difference of two cubes; Factor and Remainder Theorems; circle equation $(x-a)^2 + (y-b)^2 = r^2$; perpendicular-gradient rule (with the vertical/horizontal caveat); log product, quotient and power laws; and proof structures. Regular-polygon bearings are predecessor/ESAT material, not an Edexcel P2 topic.

4. Source manifest

Field	Value
Final-guide PDF	`guide_v2/revision_guide.pdf`
Final-guide SHA-256	`d1c96c58d9e672a52cac4c81cfc82188e36833f99081ec98da6ddb7f3fa9515f`
Content authority	Final guide only; official mathematics and lesson-status labels as carried there
Status-sensitive protected items	NSAA Q19, NSAA Q20, P2 Q3, P2 Q8(ii)
Publication state	Local-only derivative; no external export or upload

5. Quick self-check

- Can you distinguish **factor** $\Rightarrow f(c) = 0$ from **remainder** $\Rightarrow f(c)$?
- Can you write the Q6 point-slope equation through $(5, 4)$ without swapping coordinates?
- Can you state why Q7 rejects 20 using the strict log domain?
- Can you state exactly where Q8(ii) stopped live, without promoting the official proof to live work?
- Can you say which geometric-series formulae are supplied rather than memorise-only?